

ODD Protocol Description for the Agent-Based Model

Generative AI as an Interaction Intermediary: Collaboration, Trust, and Social Learning

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Model documentation prepared for repository-based reproducibility

Abstract

This document provides an ODD-style description of the agent-based model used in the study *Generative AI as an Interaction Intermediary: Modeling How AI-Mediated Interaction Changes Collaboration, Trust, and Social Learning*. The model represents a population of agents who repeatedly solve abstract collaborative tasks through one of three interaction modes: peer-first, AI-first, or hybrid interaction. The purpose of the model is exploratory and mechanism-based: it examines how repeated micro-level interaction choices can generate macro-level differences in knowledge accumulation, trust trajectories, task quality, human-interaction retention, and performance inequality. The description follows the Overview, Design concepts, and Details structure of the ODD protocol [4, 5, 6] and adopts additional decision-related clarification inspired by ODD+D [11]. It is intended to complement the main conference paper and the accompanying Python implementation.

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1 Purpose of this model document

This document is a repository-level model description. It is not intended to replace the shorter methodology section of the conference paper. Instead, it provides the additional model transparency that is difficult to include in a page-limited proceedings manuscript. The document has four purposes:

1. to describe the agent-based model using a standard ODD-style structure;
2. to make the decision rules, state variables, parameters, and update mechanisms explicit;
3. to clarify how abstract constructs such as knowledge, trust, and interaction quality are operationalised in the model;
4. to support reproducibility together with the Python code and result files in the public repository.

The model is implemented in Python and can be reproduced by running `python run_simulation.py`. The repository contains the core model file, experiment definitions, plotting scripts, generated CSV outputs, and figures. The repository URL should be inserted here when available: <https://github.com/yakupturgut/generative-ai-interaction-abm>.

2 Model scope and interpretation

The model is a compact exploratory ABM. Its purpose is not to predict the behaviour of a specific empirical population, organisation, classroom, or platform. Instead, it formalises a set of transparent assumptions about how generative AI may alter repeated interaction choices and examines their aggregate consequences. This follows a generative social-simulation logic in which system-level patterns are “grown” from local interaction rules [3]. Recent ABM scholarship similarly stresses the value of interpretable mechanisms, clear modelling assumptions, and careful validation when simulating AI-related social systems [12, 1].

The constructs used in the model should be interpreted operationally. For example, *knowledge* is not a claim that human knowledge is intrinsically one-dimensional. In the model, it denotes task-relevant capability within the simulated task environment. Similarly, *trust* represents an agent’s current propensity to rely on AI or peer interaction after repeated outcomes. These variables are represented on normalised scales because the aim is to compare interaction regimes in a transparent simulation setting.

The model uses mode-specific behavioural propensity scores. The main paper refers to these as utility functions for compactness, but they are better understood as bounded-rational choice propensities rather than empirically estimated welfare utilities. This is consistent with the use of random-utility and discrete-choice ideas for representing probabilistic choice under unobserved heterogeneity [10], while also acknowledging that agents are boundedly rational rather than fully optimizing actors [14]. The coefficients are not calibrated estimates. They encode directional assumptions that can be inspected, modified, and tested through sensitivity analysis.

3 Overview

This section gives the high-level model description required by the ODD protocol. It specifies what the model is designed to explain, which entities exist in the model, which state variables are tracked, and how processes are scheduled during each simulation step. The aim is to make the overall model logic understandable before introducing the more detailed decision and update mechanisms.

3.1 Purpose

The model examines how generative AI, when available as an interaction intermediary, can alter collaboration, trust, and social learning over repeated task episodes. The central question is not simply whether AI improves individual task performance. Rather, the model asks how different AI-use regimes change the balance between human-to-human interaction, AI-mediated support, hybrid verification, cumulative knowledge, and inequality.

The model compares four stylised regimes:

1. **Baseline human only:** AI is effectively unavailable, so interaction remains primarily peer-centred.
2. **AI as substitute:** AI is convenient and is often used in place of peer consultation.
3. **AI as complement:** AI is available but is used with verification and peer interaction.
4. **Overreliance fragility:** AI is highly convenient and trusted despite lower reliability, creating a fragile form of AI dependence.

The model outputs are used to compare these regimes in terms of mean knowledge, AI trust, peer trust, mean task quality, interaction-mode composition, Human Interaction Retention Index (HIRI), and performance inequality.

3.2 Entities, state variables, and scales

The model contains human agents connected through a weighted social network. There is no spatial grid and no physical environment. Each agent faces repeated collaborative problem-solving episodes.

Table 1: Main agent-level variables.

Variable	Interpretation	Scale / update role
k_i	Task-relevant knowledge or capability of agent i . It is an operational measure of accumulated ability in the simulated task environment.	Normalised to $[0, 1]$; increases through task quality and learning rates.
c_i	Confidence of agent i . Higher confidence increases AI-first propensity.	Normalised to $[0, 1]$; initial heterogeneous attribute.

Variable	Interpretation	Scale / update role
t_i^{AI}	Trust in AI-mediated interaction.	Normalised to $[0, 1]$; updated after AI-first and hybrid interactions.
t_i^P	Trust in peer interaction.	Normalised to $[0, 1]$; updated after peer-first and hybrid interactions.
a_i	AI literacy. This represents the capacity to make productive use of AI output.	Normalised to $[0, 1]$; affects AI-first and hybrid propensity and AI-first learning.
s_i	Sociability. This represents orientation toward peer consultation.	Normalised to $[0, 1]$; affects peer-first and hybrid propensity.
v_i	Verification tendency. This represents the tendency to check, combine, or scrutinise information.	Normalised to $[0, 1]$; affects AI-first difficulty penalty and hybrid propensity/quality.
m_i	Recent success memory.	Normalised to $[0, 1]$; exponentially smoothed from task success.
P_i	Cumulative performance.	Non-negative cumulative sum of realised task quality.

The network contains weighted undirected edges. Edge weights represent social tie strength. They are reinforced when peer or hybrid interaction succeeds and decay when ties remain unused.

Table 2: Main global and scenario-level variables.

Variable	Interpretation	Scale / value
n	Number of agents.	120
T	Number of task episodes.	60
d_t	Difficulty of task episode t .	Uniform draw from $[0.30, 0.90]$
ϕ_t	AI reliability in a scenario.	Scenario parameter
κ	AI convenience in a scenario.	Scenario parameter
ψ	Verification strength.	Scenario parameter
w_{ij}	Tie strength between connected agents i and j .	Initialised in $[0.8, 1.2]$ and updated dynamically

3.3 Process overview and scheduling

Each simulation step represents one abstract collaborative problem-solving episode, not a calendar unit. At each step, all agents face the same difficulty draw. The scheduling is synchronous at the level of aggregate task episodes but updates are applied agent-wise within each step.

At each step:

1. Draw task difficulty $d_t \sim U(0.30, 0.90)$.
2. Compute each agent’s behavioural propensity for peer-first, AI-first, and hybrid interaction.
3. Convert propensities into probabilities using softmax.
4. Sample each agent’s interaction mode.
5. For peer-first and hybrid agents, select one local network neighbour.
6. Compute task quality for each agent using the selected mode-specific quality function.
7. Update knowledge, trust, recent success memory, cumulative performance, and social tie weights.
8. Record aggregate outputs.

4 Design concepts

This section explains the conceptual assumptions behind the model. It clarifies what is expected to emerge from the simulation, how agents adapt, how they make interaction choices, what they can observe, and where stochasticity enters the model. These design concepts are especially important because the model is exploratory and mechanism-based rather than empirically calibrated.

4.1 Basic principles

The model is built around the idea that generative AI can act as an intermediary in social interaction. In many collaborative settings, AI does not merely provide information; it can change whom people consult, whether they verify information with peers, and how much human interaction remains over time. The model therefore distinguishes between three modes of interaction:

- **Peer-first interaction:** the agent primarily consults another human agent.
- **AI-first interaction:** the agent primarily consults AI.
- **Hybrid interaction:** the agent combines AI use with peer consultation.

This design allows the model to represent both augmentation and substitution. AI may complement human interaction when it is combined with verification and peer learning, or it may substitute for human interaction when convenience and trust make AI-first interaction dominant.

4.2 Emergence

The main outcomes are emergent in the sense that they are not imposed directly. For example, the model does not force peer trust to decline or AI trust to increase. These trajectories emerge

from repeated task outcomes and trust updating. Similarly, the Human Interaction Retention Index (HIRI) emerges from the number of peer-first and hybrid interactions relative to the initial human-interaction level.

4.3 Adaptation

Agents adapt through three mechanisms:

1. **Knowledge accumulation:** task quality increases task-relevant knowledge.
2. **Trust updating:** successful interactions increase trust in the used source, while unsuccessful interactions reduce trust.
3. **Network reinforcement:** successful peer or hybrid interactions strengthen the corresponding tie, while unused ties decay.

These mechanisms create feedback between interaction choices and future interaction opportunities. For example, repeated AI-first interaction may reduce the use of peer ties, which may gradually thin the human interaction structure.

4.4 Objectives and decision-making

Agents do not solve a fully rational optimisation problem. Instead, they select an interaction mode probabilistically according to mode-specific behavioural propensities. These propensities combine agent attributes, trust, local social cues, AI convenience, verification tendency, and task difficulty.

For agent i , the propensity of peer-first interaction is:

$$U_i(P) = 1.15s_i + 1.00t_i^P + 0.70\bar{m}_{N_i} + 0.20\bar{w}_{N_i} - 0.65d_t, \quad (1)$$

where s_i is sociability, t_i^P is peer trust, $\bar{m}_{N_i}$ is the average recent success of neighbours, $\bar{w}_{N_i}$ is the average local tie strength, and d_t is task difficulty.

The propensity of AI-first interaction is:

$$U_i(A) = 1.15a_i + 0.95t_i^{AI} + 0.55c_i + 0.75\kappa - 0.55d_t(0.4 + v_i\psi), \quad (2)$$

where a_i is AI literacy, t_i^{AI} is trust in AI, c_i is confidence, κ is AI convenience, v_i is verification tendency, and ψ is verification strength.

The propensity of hybrid interaction is:

$$U_i(H) = 0.95a_i + 0.85s_i + 0.60(t_i^{AI} + t_i^P) + 0.45v_i + 0.35\kappa - 0.42d_t. \quad (3)$$

The probability of selecting interaction mode $x \in \{P, A, H\}$ is:

$$Pr_i(x) = \frac{\exp(U_i(x))}{\exp(U_i(P)) + \exp(U_i(A)) + \exp(U_i(H))}. \quad (4)$$

The use of a softmax rule ensures that choice remains probabilistic. Even when one mode has a higher propensity, the agent may still choose another mode. This reflects unobserved heterogeneity, noise, and bounded rationality.

4.5 Learning

Learning is represented as task-relevant capability growth. Peer-first and hybrid interaction produce stronger learning increments than AI-first interaction. In the AI-first case, learning is additionally moderated by AI literacy, reflecting the assumption that direct AI use does not automatically translate into durable learning without sufficient capability to interpret and use AI output.

4.6 Prediction

Agents do not explicitly forecast the future. Their choices are based on current attributes, current trust, recent neighbour success, local tie strength, AI convenience, task difficulty, and verification tendency.

4.7 Sensing

Agents sense their own attributes, their own trust states, the current task difficulty, and local network cues. Peer-first propensity uses the average recent success and tie strength of neighbours. Agents do not observe the full global state of the system.

4.8 Interaction

Interaction occurs through peer consultation, AI consultation, or hybrid consultation. Peer-first and hybrid agents select one neighbour from their local network. AI-first agents do not use a peer in that step. Hybrid agents use both AI and peer input.

4.9 Stochasticity

Stochasticity enters through:

- initial heterogeneous agent attributes;
- network construction and initial tie weights;
- task difficulty draws;
- probabilistic mode choice;
- random peer selection among neighbours;
- task-quality noise.

All stochastic processes are controlled by explicit random seeds in the experiment scripts.

4.10 Collectives

The main collective structure is the social network. There are no formal teams or organisations. However, local neighbourhoods act as the relevant interaction context for peer-first and hybrid modes.

4.11 Observation

The following outputs are recorded at each step:

- mean task quality;
- mean knowledge;
- mean trust in AI;
- mean trust in peers;
- cumulative mean performance;
- number of peer-first, AI-first, and hybrid events;
- success rate;
- average tie strength;
- Human Interaction Retention Index (HIRI);
- AI-mediated reliance rate (AMR);
- Gini coefficient of cumulative performance.

5 Details

This section reports the concrete implementation details that allow the model to be inspected and reproduced. It describes how agents and the network are initialized, which external inputs are used, how the submodels operate, and how outputs are measured. These details correspond to the computational implementation in the accompanying Python files.

5.1 Initialization

The model is initialized with $n = 120$ agents and $T = 60$ task episodes. Initial agent attributes are drawn from bounded normal distributions:

Table 3: Initial agent attribute distributions.

Variable	Distribution before clipping	Clipping interval
Knowledge k_i	$N(0.52, 0.12^2)$	$[0.10, 0.95]$
Confidence c_i	$N(0.50, 0.15^2)$	$[0.05, 0.95]$
AI trust t_i^{AI}	$N(0.52, 0.16^2)$	$[0.05, 0.95]$

Variable	Distribution before clipping	Clipping interval
Peer trust t_i^P	$N(0.58, 0.14^2)$	$[0.05, 0.95]$
AI literacy a_i	$N(0.48, 0.18^2)$	$[0.05, 0.95]$
Sociability s_i	$N(0.55, 0.15^2)$	$[0.05, 0.95]$
Verification v_i	$N(0.55, 0.17^2)$	$[0.05, 0.95]$

The default network is a connected Watts-Strogatz small-world graph. Small-world networks are used because they represent local clustering together with some long-range reach, a common abstraction for social interaction structures [15]. With 120 agents and network density 0.06, each agent has a limited local neighbourhood rather than full access to the entire population. The population size is also within the broad range often discussed in relation to human social network size, while recognising that Dunbar-type numbers are debated and should not be treated as a fixed universal constant [7, 9].

5.2 Input data

The model does not use external empirical input data. Scenario settings are defined directly in `experiments.py`. The absence of empirical calibration is intentional: the model is a mechanism-based exploratory model. The parameter values define stylised regimes rather than fitted estimates.

Table 4: Scenario parameters.

Scenario	ϕ AI reliability	κ convenience	ψ verification	Learning rates (P, A, H)	
Baseline human only	0.00	-2.00	0.60	0.042, 0.000	0.000,
AI as substitute	0.80	0.95	0.25	0.038, 0.028	0.022,
AI as complement	0.82	0.72	0.80	0.042, 0.040	0.018,
Overreliance fragility	0.65	1.10	0.12	0.034, 0.022	0.024,

5.3 Submodels

5.3.1 Task difficulty

At each step, a task difficulty value is drawn as:

$$d_t \sim U(0.30, 0.90). \quad (5)$$

The task is abstract by design. It represents a repeated collaborative problem-solving episode rather than a domain-specific task. This abstraction allows the model to compare interaction regimes without tying the results to a specific classroom, workplace, or platform.

5.3.2 Peer selection

If an agent chooses peer-first or hybrid interaction, one peer is selected randomly from the agent's local network neighbours. If an agent chooses AI-first interaction, no peer is selected in that step.

5.3.3 Task quality

Task quality is computed differently for each mode. For a selected peer $j \in N_i$, peer-first quality is:

$$q_i^P = [0.44k_i + 0.30k_j + 0.12t_i^P + 0.10v_i - 0.40d_t + \epsilon_i]_0^1. \quad (6)$$

AI-first quality is:

$$q_i^A = [0.34a_i + 0.26t_i^{AI} + 0.30\phi_t + 0.08c_i - 0.34d_t - 0.08v_i(1 - \phi_t) + \epsilon_i]_0^1. \quad (7)$$

Hybrid quality is:

$$q_i^H = [0.22a_i + 0.22\phi_t + 0.18k_j + 0.14v_i + 0.10t_i^P + 0.08t_i^{AI} - 0.28d_t + \epsilon_i]_0^1. \quad (8)$$

Here, $\epsilon_i \sim N(0, 0.045^2)$ and $[\cdot]_0^1$ denotes clipping to $[0, 1]$.

5.3.4 Knowledge updating

Knowledge is updated after each task episode. For peer-first agents:

$$k_i^{t+1} = k_i^t + \lambda_P q_i^t. \quad (9)$$

For AI-first agents:

$$k_i^{t+1} = k_i^t + \lambda_A a_i q_i^t. \quad (10)$$

For hybrid agents:

$$k_i^{t+1} = k_i^t + \lambda_H q_i^t. \quad (11)$$

Knowledge is clipped to $[0, 1]$ after updating.

5.3.5 Trust updating

A task is considered successful when $q_i^t \geq 0.50$. The trust update is:

$$\Delta t_i^z = \begin{cases} \mu^+, & \text{if the interaction using source } z \text{ succeeds,} \\ -\mu^-, & \text{otherwise,} \end{cases} \quad (12)$$

where $z \in \{AI, P\}$. AI-first interaction updates AI trust. Peer-first interaction updates peer trust. Hybrid interaction updates both AI trust and peer trust. Trust is clipped to $[0, 1]$.

5.3.6 Recent success memory

Recent success memory is updated as:

$$m_i^{t+1} = 0.70m_i^t + 0.30\mathbb{I}(q_i^t \geq 0.50). \quad (13)$$

This creates a short-term memory of performance that can influence future peer-first propensity through local neighbour success.

5.3.7 Tie updating

If agent i uses peer j in peer-first or hybrid interaction, the tie weight w_{ij} is reinforced after success and weakened after failure. Unused ties decay after the specified decay interval. This mechanism allows AI substitution to affect not only immediate task quality but also the longer-run strength of human interaction pathways.

5.4 Output measures

The model records both performance-related and interaction-related outputs. The Human Interaction Retention Index is defined as:

$$HIRI_t = \frac{\text{peer-first events}_t + \text{hybrid events}_t}{\text{peer-first events}_1 + \text{hybrid events}_1}. \quad (14)$$

Values closer to one indicate stronger retention of human-involving interaction relative to the initial step. Lower values indicate greater substitution away from human-involving interaction.

Performance inequality is measured using the Gini coefficient of cumulative performance:

$$G = \frac{\sum_{i=1}^n \sum_{j=1}^n |P_i - P_j|}{2n \sum_{i=1}^n P_i}. \quad (15)$$

6 Simulation experiments

This section describes how the model is used to generate the results reported in the main paper. The simulation experiments are designed to compare four stylised AI-use regimes and to examine whether the qualitative findings are sensitive to selected parameter changes. The same experiment definitions are implemented in `experiments.py`, while `run_simulation.py` runs the full workflow and exports the results.

6.1 Main scenario experiment

The main experiment runs four scenarios with 20 replications per scenario. Seeds start at 2026 and are offset by scenario and replication. The main outputs are saved as:

- `replication_histories.csv`;
- `final_outcomes_by_replication.csv`;
- `aggregated_results.csv`;
- `summary_means.csv`.

6.2 Sensitivity analysis

A one-at-a-time sensitivity analysis is conducted around the AI-as-complement scenario. The varied factors are AI reliability, AI convenience, verification strength, and network density. Each factor is evaluated at low, base, and high levels with 6 replications per level. The sensitivity output is saved as `sensitivity_results.csv`.

7 Implementation and reproducibility

This section explains how the computational experiment can be recreated from the public repository. The model is distributed together with source code, generated outputs, figures, and package requirements so that readers can inspect both the conceptual model and the executable implementation. The workflow is intentionally script-based rather than notebook-based to make the experiment easier to rerun from a clean environment.

The implementation is written in Python. The required packages are:

- `numpy`;
- `pandas`;
- `networkx`;
- `matplotlib`.

The full pipeline can be recreated from the command line:

```
pip install -r requirements.txt
python run_simulation.py
```

The script regenerates the scenario outputs, sensitivity-analysis outputs, and all PNG/PDF figures. No Jupyter notebook is required; the experiment is fully scripted.

8 Model boundaries and limitations

The model is intentionally simple. Its boundaries are important for interpretation:

- The model is not calibrated to a specific empirical population.
- Knowledge, trust, sociability, AI literacy, and verification tendency are operational constructs.
- The task is abstract and represents a repeated collaborative problem-solving episode.
- The small-world network is a stylised representation, not an empirical social network.
- Utility functions are behavioural propensity functions, not estimated welfare utilities.
- Scenario parameters encode stylised AI-use regimes rather than measured institutional settings.

These limitations are also useful because they make the model interpretable. Future work can replace the abstract task with domain-specific tasks, calibrate parameters from experiments, compare alternative network structures, and test alternative decision rules.

9 References

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